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 Studierichting: Informatica
 Oefeningenreeks 2E nr. 10.9

$$\begin{aligned}
 8(16^{x-1} - 4^{2x-1}) &= 3(2^{x+2} - 4^x - 2 \cdot 8^x) \\
 \Leftrightarrow 2^3 (2^{4x-4} - 2^{4x-2}) &= 3 (2^{x+2} - 2^{2x} - 2^{3x+1}) \\
 \Leftrightarrow 2^{4x-1} - 2^{4x+1} &= 3 (2^{x+2} - 2^{2x} - 2^{3x+1})
 \end{aligned}$$

Stel $t = 2^x$

$$\begin{aligned}
 \frac{1}{2}t^4 - 2t^4 &= 3(4t - t^2 - 2t^3) \\
 \Leftrightarrow \frac{-3}{2}t^4 + 6t^3 + 3t^2 - 12t &= 0 \\
 \Leftrightarrow t = 0 \text{ of } \frac{-3}{2}t^3 + 6t^2 + 3t - 12 &= 0
 \end{aligned}$$

Oplossen 3de-graadsvergelijking:

$$\begin{array}{c|ccc}
 4 & \frac{-3}{2} & 6 & 3 & -12 \\
 & & -6 & 0 & 12 \\
 \hline
 & \frac{-3}{2} & 0 & 3 & 0 \\
 \hline
 \frac{-3}{2}t^2 + 3 & = & 0 \\
 \Leftrightarrow t^2 & = & 2
 \end{array}$$

Alle oplossingen van 4de-graadsvergelijking:

$$\begin{aligned}
 t = 0 \text{ of } t = 4 \text{ of } t = +\sqrt{2} \text{ of } t = -\sqrt{2} \\
 \Leftrightarrow t = 2^x \\
 2^x = 0 \text{ of } 2^x = 4 \text{ of } 2^x = 2^{\frac{1}{2}} \text{ of } 2^x = -\left(2^{\frac{1}{2}}\right) \\
 \Leftrightarrow x = / \text{ of } x = 2 \text{ of } x = \frac{1}{2} \text{ of } x = /
 \end{aligned}$$

References